

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: GEN-302 — Introduction to IoT

Time: Three hours

Maximum Marks: 70

Instructions: Answer all Sections. Candidates are required to write their answers in their own words as far as practicable.

SECTION — A: Long Answer Questions

[15 × 2 = 30]

Instructions: Long answer type questions to be answered in about 750 words each. Attempt any two.

Question 1. What is the structure of an Arduino sketch? Explain with a suitable example.

OR

What are the major components of the Arduino Uno Microcontroller? Briefly explain.

Question 2. How can you develop an IoT-based project for a smart agriculture system?

OR

How can you develop an IoT-based project for a smart city?

SECTION — B: Short Answer Questions

[5 × 6 = 30]

Instructions: Short answer type questions to be answered in about 450-550 words each. Attempt any six.

Question 1. What are the main components of the IoT architecture?

Question 2. What are the primary hardware components that make up an embedded system?

Question 3. What are some examples of sensors that can be used in home automation?

Question 4. What are some of the protocols used for IoT communication?

Question 5. What are the differences between IPv4 and IPv6?

Question 6. How could 5G cellular networks impact IoT?

Question 7. What steps can an organization take to protect IoT systems and devices?

Question 8. Discuss different types of Drones and their functionality.

SECTION — C: Multiple Choice Questions

[10 × 1 = 10]

Instructions: Select the correct option for each question.

Question 1. Which of the following IoT networks has a very short range?

- (a) Short Network
- (b) LPWAN
- (c) SigFox
- (d) Short Range Wireless Network

Question 2. What is the main purpose of WoT (Web of Things) in the IoT?

- (a) Improve usability and interoperability
- (b) Reduce security
- (c) Make development complex
- (d) Increase the cost

Question 3. Which of the following is not an application of IoT?

- (a) Wearables
- (b) Smart Grid
- (c) Arduino
- (d) Smart City

Question 4. What is the full form of MQTT?

- (a) Multi-Queue Telemetry Things
- (b) Multiple Queue Telemetry Things
- (c) Message Queue Telemetry Things
- (d) Message Queue Telemetry Transport

Question 5. Which of the following layers provides end-to-end communication in IoT?

- (a) Logical Layer
- (b) Data Link Layer
- (c) Transport Layer
- (d) Session Layer

Question 6. Which of the following devices is used to measure gases or liquids?

- (a) Optical Sensor
- (b) Gas Sensor
- (c) Smoke Sensor
- (d) Pressure Sensor

Question 7. Which of the following topologies is used for ZigBee Smart Energy?

- (a) Bus Topology
- (b) Ring Topology
- (c) Star Topology
- (d) Any Topology

Question 8. Identify the protocol used to link all devices in IoT:

- (a) TCP/IP
- (b) UDP
- (c) Network
- (d) HTTP

Question 9. In IoT, as one of the key characteristics, devices have different hardware platforms and networks. This is known as:

- (a) Sensors
- (b) Heterogeneity
- (c) Security
- (d) Connectivity

Question 10. Which of the following issues are considered in IoT?

- (a) Security Issue
- (b) Reliability Issue
- (c) Standard Issue
- (d) All of the above